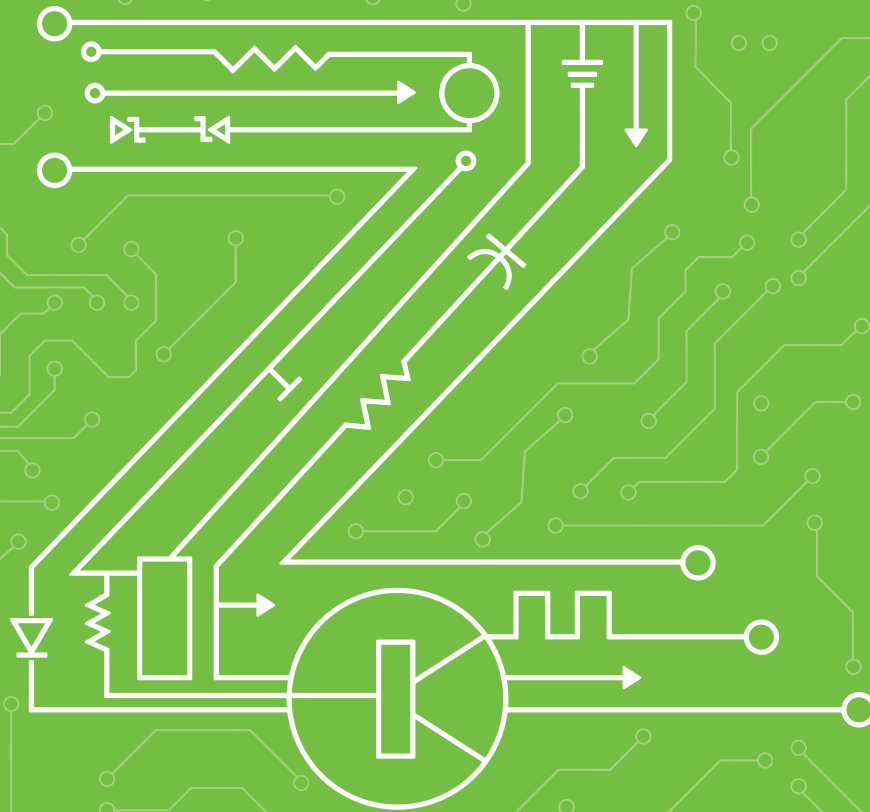




EASY AS PI

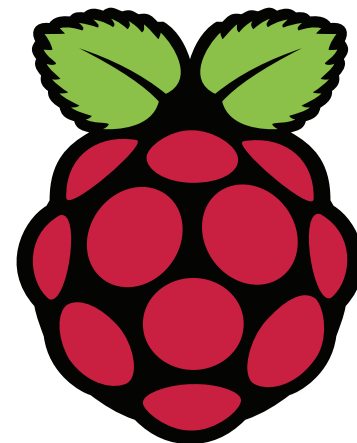
YOUR RASPBERRY PI PROJECT GUIDE



WELCOME

You're here because you know that the best things in life are worth tinkering with, reimagining, and doing yourself. We think so, too.

That's why we've teamed up with Raspberry Pi to bring 314 new perspectives (like yours) to a growing community of digital makers. We want you to explore what drives you to create—but that doesn't mean you have to go it alone. So we put together some of our favorite Pi projects to help you get started (along with some extra ingredients we thought you might like). Let's go!



INSIDE YOUR KIT

We want to make sure you hit the ground running, so we've included:

- 1 Raspberry Pi Zero W
- 1 microSD card
- 1 microSD to SD card adapter
- 1 micro USB cable
- 1 micro USB to USB adapter
- 1 micro USB AC power plug
- 1 mini HDMI to HDMI adapter

ADDITIONAL INGREDIENTS

If you're feeling adventurous, you might want to buy the following:

- USB Hub
- HDMI display or Raspberry Pi Screen
- soldering iron and solder
- wire and wire stripper
- board pins (2x20)

SAFETY FIRST!



If you solder, do so in a well-ventilated space.



Wear safety glasses when soldering or using power tools.

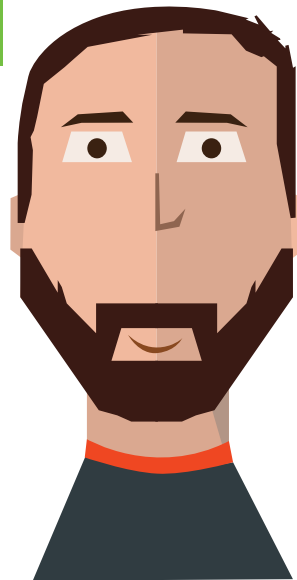
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Retro Game Console
by Jon Prusik

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RPI Docker
by James Scheffler

p. 11



Panda Cam
by Teenie Rose

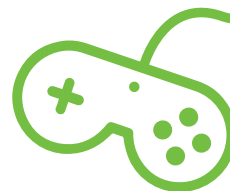
p. 16



MIDI Synth
by Alessandro Rossi

RETROPIE GAME CONSOLE

by Jon Prusik



Using RetroPie, an operating system dedicated to retro gaming on the Raspberry Pi, you can assemble a portable console for on-the-go gaming. If you get stuck at any point, RetroPie also has a guide right [here](#).



1 HOUR



EASY

PHYSICAL INGREDIENTS:

- 1 Raspberry Pi Zero
- 1 HDMI cable
- 1 HDMI display (most modern televisions have an HDMI port)
- 1 mini HDMI to HDMI adapter
- 1 micro USB AC power plug
- 1 micro USB cable
- 1 micro USB to USB adapter
- 1 USB controller (I use a Xbox 360 Controller with the wireless PC USB adapter)
- 1 keyboard (optional, but recommended for setup)
- 1 USB hub (optional)
- 1 USB wireless adapter (optional)

DIGITAL INGREDIENTS:

- [RetroPie for Raspberry Pi Zero](#) (double click link to download)
- [Win32DiskImager](#) (for Windows)
- [ApplePi Baker](#) (for Mac)

KNOWLEDGE NEEDED:

- How to download and manipulate files on your computer
- How to extract .gz files on your computer
- How to configure your WiFi access point (optional)

RECIPE



STEP 1: DOWNLOAD AND PREPARE RETROPIE

- Visit [RetroPie](#) and click the download button for “Raspberry* Pi 0/1.”
- Extract the file you just downloaded (ends with “.gz”) and you should end up with an image file (ends with “.img”). Save the image file some where you’ll remember.
- Download and install [Win32DiskImager](#) (for Windows) or [ApplePi Baker](#) (for Mac).

STEP 2: INSTALL RETROPIE

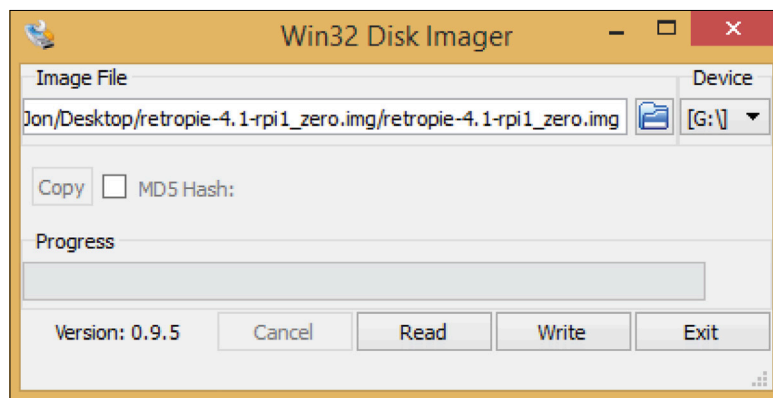
IMPORTANT: In the following steps, be careful to only select your SD card drive—all the data on the selected drive will be overwritten and lost!

Insert your microSD card (via SD adapter if needed) into your computer.

FOR WINDOWS:

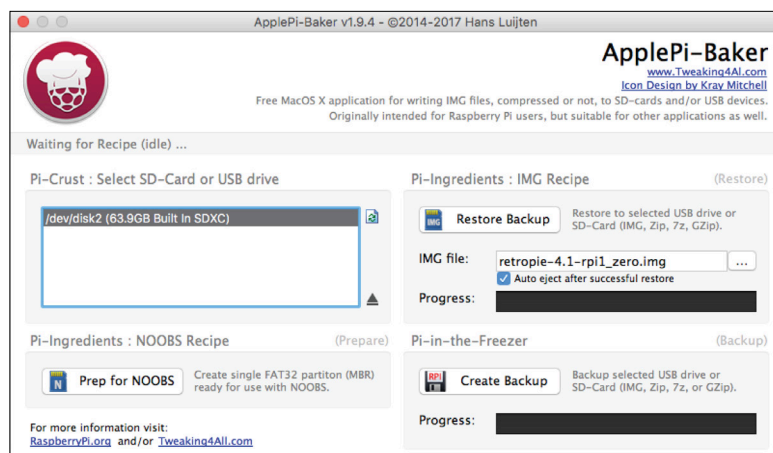
Open Win32 Disk Imager and select the image file you downloaded.

Under “Device,” make sure the drive for your SD card is selected and click the “write” button.



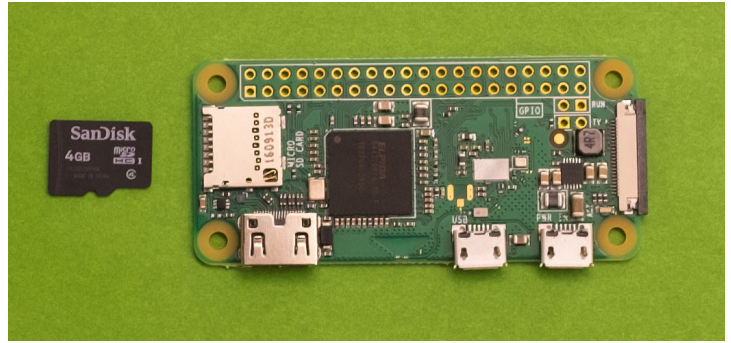
FOR MAC:

Open ApplePi Baker and look for the “Pi-Crust” section. Make sure your SD card is selected. Now look for the “Pi-Ingredients” section. Load the image file you downloaded, and click the “Restore Backup” button.

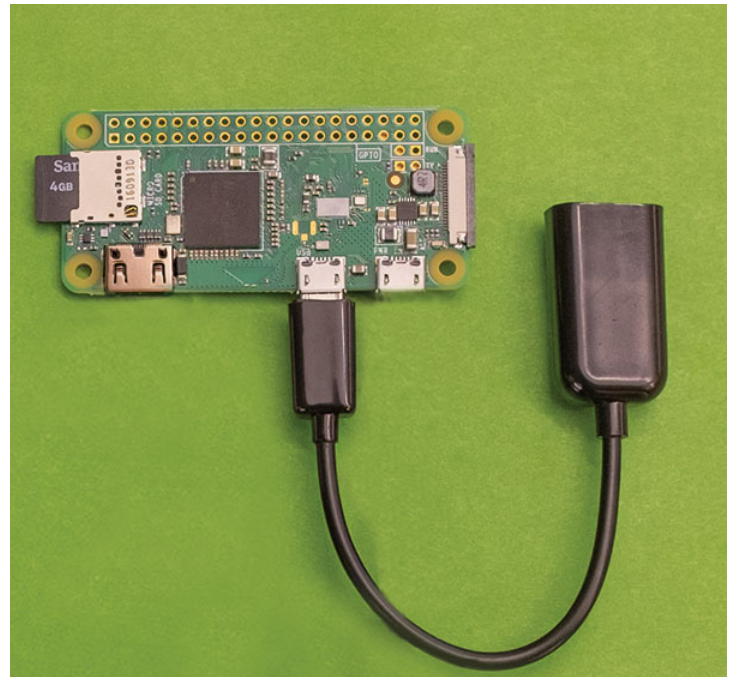


STEP 3: PLUGGING IN YOUR CABLES AND POWER ON RETROPIE

Insert your microSD card into your Pi Zero.



Plug your micro USB to USB adapter into the micro USB port on the Pi Zero labeled "USB."



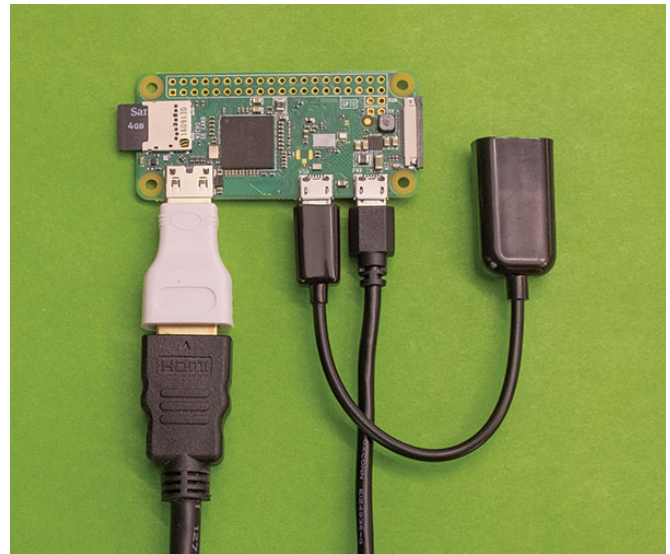
(optional) If you have a USB hub that you'd like to use, plug it into the USB adapter now.

Note: If you have a USB wireless adapter and would like to use it to give your console internet capabilities, you'll need a USB hub so that you can plug in the wireless adapter at the same time as your controller and/or keyboard.



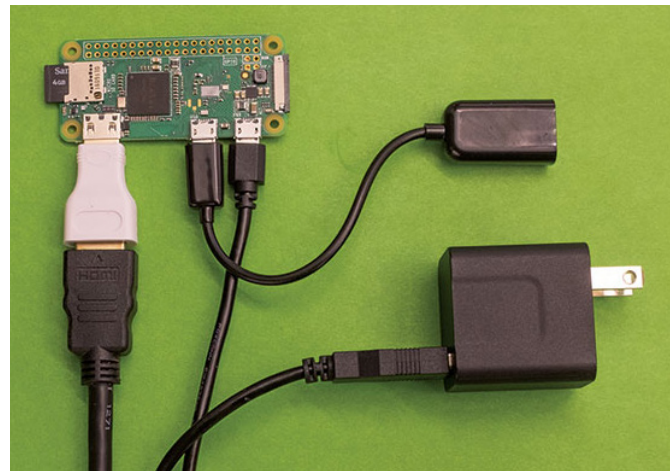
(optional) If you have a keyboard you'd like to use, plug it into the USB hub or adapter now. Otherwise, plug your USB controller into the USB hub or adapter.

Plug in your mini HDMI to HDMI adapter into the port on the Pi Zero labeled "HDMI."



Connect your Pi to the HDMI port of your HDMI display with an HDMI cable.

This next step will actually power on and boot the Raspberry Pi as soon as you plug in, so double check that everything is plugged in and your display is set to the correct HDMI input.



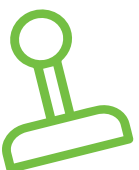
Plug in the micro USB AC power plug into the micro USB port on the Pi Zero labeled "PWR IN."

After a few seconds, you should see the RetroPie load screen.



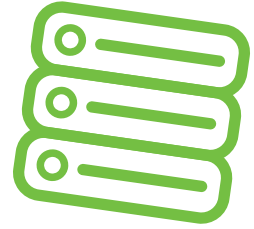
STEP 4 - CONFIGURING RETROPIE

From here on out, RetroPie will guide you through configuring your system. It will ask to assign controller buttons, and make the menu available for all kinds of setting changes.



RPI DOCKER

by James Scheffler



[Docker](#) containers are an easy way to install and run applications without having to worry about dependencies, prerequisites, or interference with other services on a system. It's for installing server-based things like [WordPress](#) or [MySQL](#)—and now you can try it out on your Raspberry Pi. Using [HypriotOS](#), a Linux distribution for running Docker containers on the Raspberry Pi, you can create as many Docker hosts as you want!



1 HOUR



MEDIUM

PHYSICAL INGREDIENTS:

- 1 Raspberry Pi Zero W
- 1 micro SD card
- 1 micro USB AC power plug
- 1 micro USB cable

DIGITAL INGREDIENTS:

- A copy of [HypriotOS](#)
- An SSH client. On Mac, you already have one by accessing your terminal. On Windows, you need a program like [puTTY](#).
- [Win32DiskImager](#) (for Windows)
- [ApplePi Baker](#) (for Mac)

KNOWLEDGE NEEDED:

- How to download and manipulate files on your computer
- How to extract .zip files on your computer
- How to configure your WiFi access point

RECIPE



STEP 1: DOWNLOAD AND PREPARE HYPRIOTOS

- Download the most recent HypriotOS image file that you find [here](#).
- Extract the file you just downloaded (ends with “.zip”). You should end up with an image file (ends with “.img”).
- Save the image file somewhere you you’ll remember.
- Download and install [Win32DiskImager](#) (for Windows) or [ApplePi Baker](#) (for Mac).

STEP 2: INSTALL HYPRIOTOS

IMPORTANT: In the following steps, be careful to only select your SD card drive—all the data on the selected drive will be overwritten and lost!

Insert your microSD card (via SD adapter if needed) into your computer, then use the tool of your choice to install your image onto the SD card. It isn’t enough to simply copy the file there, so you’ll need to proceed with a tool:

FOR WINDOWS:

Open Win32 Disk Imager and select the image file you downloaded. Under “Device,” make sure the drive for your SD card is selected and click the “write” button.

FOR MAC:

Open ApplePi Baker and look for the “Pi-Crust” section. Make sure your SD card is selected. Now look for the “Pi-Ingredients” section. Load the image file you downloaded, and click the “Restore Backup” button.

STEP 3: USE A PLAIN TEXT EDITOR TO CONFIGURE HYPRIOTOS FOR WIFI

Close the tool that you were using to install the image once it has completed. Remove the microSD card from your computer for a moment, then reinsert your microSD card so that your computer can see it as a regular storage device with files on it. Using your plain text editor, (e.g. notepad on Windows or TextEdit for Mac) open `device-init.yaml`.

Edit the lines under **# optional wireless network settings :**

First, delete the `#` that you see at the beginning of the three last lines, but leave the rest of the lines alone. This is called “uncommenting” by programmers. Then change `ssid: “MyNetwork”` so that your own WiFi network name is between the quotation marks. Finally, change `password: “secret_password”` so that your WiFi password is between the quotation marks.

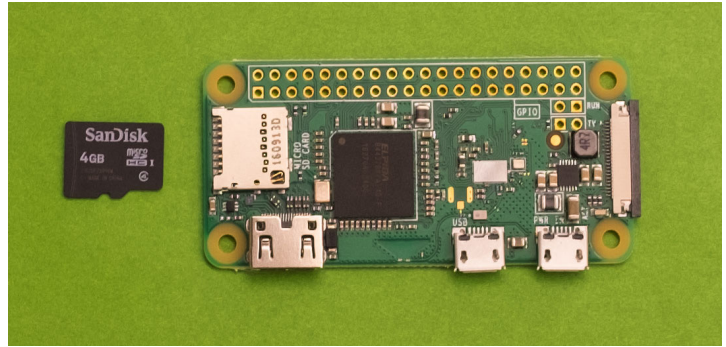
Optionally, you may like to name your Pi. Change the host name to whatever you like, but don’t forget it! Now be absolutely certain that you are saving this file as PLAIN TEXT, and safely remove your SD card from your computer.

STEP 4: PLUGGING IN YOUR CABLE AND POWER ON HYPRIOTOS

Insert your microSD card into your Pi Zero.

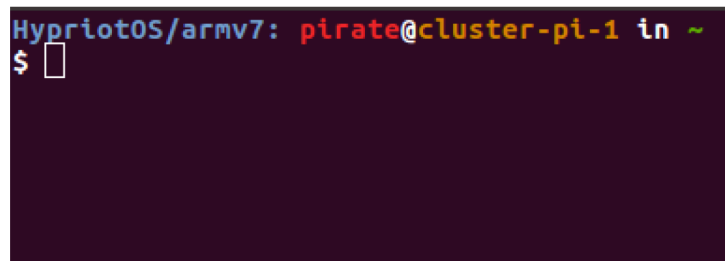
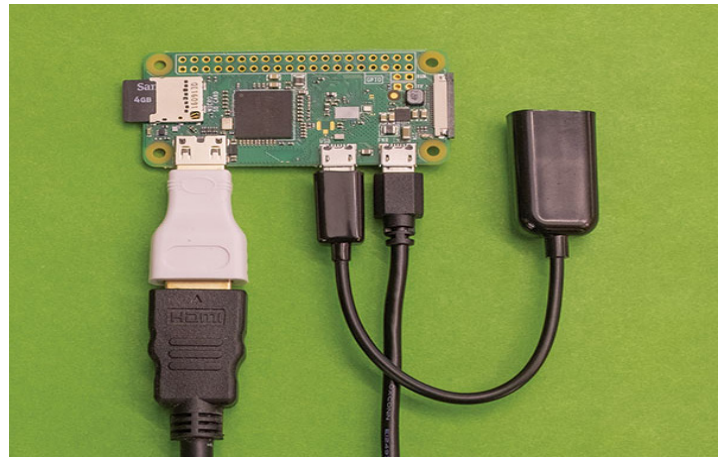
Power on and boot the Raspberry Pi by plugging in the micro USB AC power plug into the micro USB port on the Pi Zero labeled “PWR IN.”

Count to ten or so while HypriotOS boots up.



STEP 5: CONNECT TO HYPRIOTOS

Here's where your SSH client comes in. Use it to connect to your Raspberry Pi over WiFi. If you didn't change the hostname in device-init.yaml then you want to connect to the user `pirate@black-pearl.local`, otherwise use the hostname you chose: `pirate@hostname.local`.



The default password for pirate is hypriot. You should see a screen that looks like this:

STEP 6: YOUR FIRST DOCKER CONTAINER!

Let's start small, and set up a tiny webserver. At your command line prompt, type: `docker run -d -p 80:80 hypriot/rpi-busybox-httpd`

This will download and start the Docker image `hypriot/rpi-busybox-httpd`, which contains a tiny webserver. Check to see if your container is running: `docker ps`

You should see the container you just started in the container list. Now the magic happens. Open any browser you like, and navigate to the `hostname.local` or the IP address of your Raspberry Pi. You'll see a black background, with the Raspberry Pi and Docker logos side by side served up by your own Raspberry Pi webserver, running inside of a Docker container!

GOING FURTHER

Most of this guide follows the steps in [HypriotOS's Getting Started Guide](#). You Should definitely check out that documentation, and their references. But you may also want to know more about Docker specifically, and for that you'll want to see [Docker's Own Getting Started Guide](#), bearing in mind that your Raspberry Pi is an ARM architecture device, and so you'll have to choose your Docker images carefully if you want them to work correctly.



Times have seriously changed. Embedded systems like Raspberry Pi make this sort of computing easily available to anyone with a little bit of curiosity.

PANDA CAM

by Teenie Rose



Using [Raspbian's](#) startup settings, you can configure your Raspberry Pi Zero to start up and launch the web browser in fullscreen for a webpage of your choice. At Zipcar, we're fans of the San Diego Zoo's Giant Panda Cam (but you can use this tutorial to set up a monitor for any site you wish).



1 HOUR



EASY

PHYSICAL INGREDIENTS:

- 1 HDMI cable
- 1 HDMI display (most modern televisions have an HDMI port)
- 1 keyboard (needed for setup)
- 1 micro USB AC power plug
- 1 micro USB to USB adapter (needed for setup)
- 1 mini HDMI to HDMI adapter
- 1 Raspberry Pi Zero W

DIGITAL INGREDIENTS:

- [Raspbian](#)
- [Win32DiskImager](#) (for Windows)
- [ApplePi Baker](#) (for Mac)

KNOWLEDGE NEEDED:

- How to download and manipulate files on your computer
- How to extract .zip files on your computer
- Configuration of your WiFi access point

RECIPE



STEP 1: DOWNLOAD AND PREPARE RASPBIAN

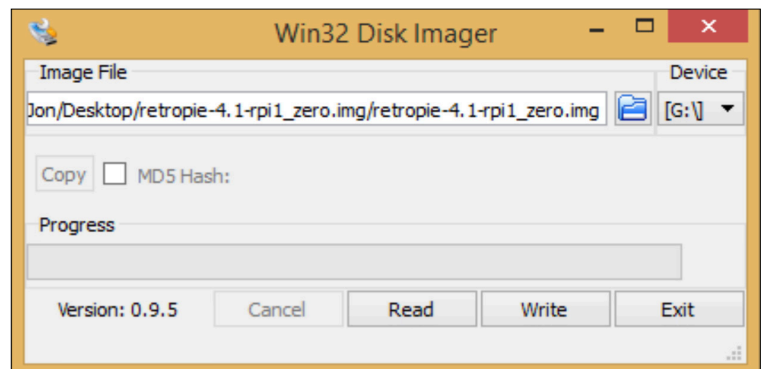
- Download [Raspbian](#).
- Extract the file you just downloaded (ends with “.zip”); you should end up with an image file (ends with “.img”). Save the image file somewhere you can get to it in the next step.
- Download and install [Win32DiskImager](#) (for Windows) or [ApplePi Baker](#) (for Mac).

STEP 2: INSTALL RASPBIAN

IMPORTANT: In the following steps, be careful to only select your SD card drive—all the data on the selected drive will be overwritten and lost! Insert your microSD card (via SD adapter if needed) into your computer.

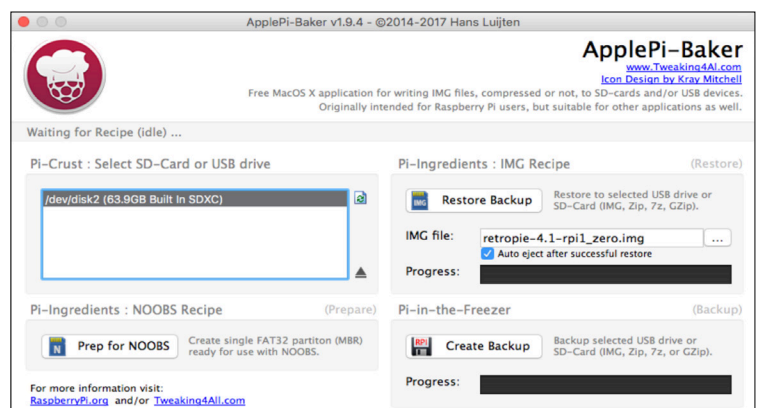
FOR WINDOWS:

- Open Win32 Disk Imager and select the image file you downloaded.
- Under “Device,” make sure the drive for your SD card is selected and click the “write” button.



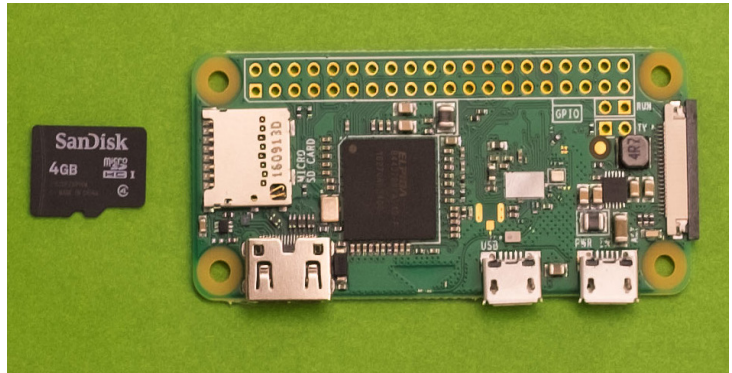
FOR MAC:

- Open ApplePi Baker and look for the “Pi-Crust” section. Make sure your SD card is selected.
- Now look for the “Pi-Ingredients” section. Load the image file you downloaded, and click the “Restore Backup” button.

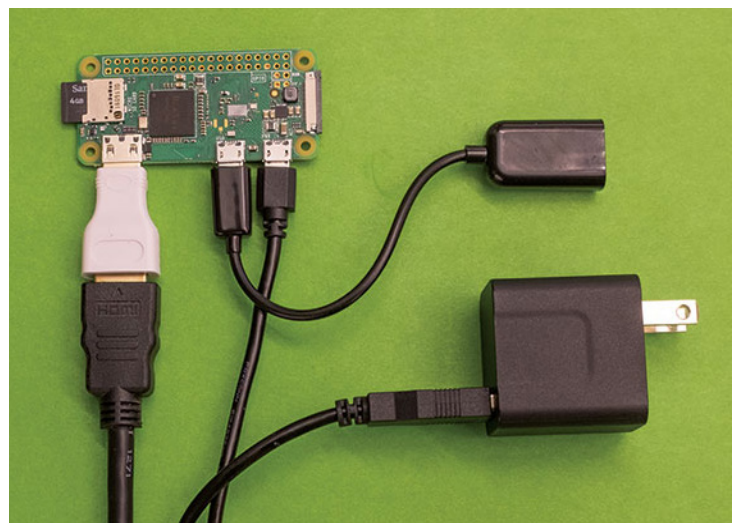
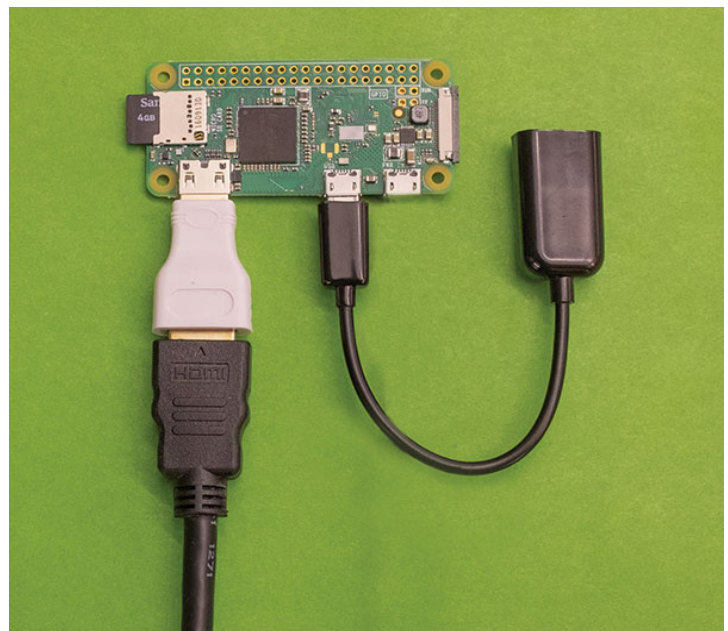


STEP 3: PLUGGING IN YOUR CABLES AND POWER ON RASPBIAN

- Insert your microSD card into your Pi Zero.



- Plug your micro USB to USB adapter into the micro USB port on the Pi Zero labeled “USB.”
- If you have a keyboard with a touchpad or wireless keyboard/mouse combo, you may want to use that for setup. If you require two USB ports for your mouse and keyboard, you’ll need to plug them into a USB hub, which you can plug into the USB adapter now. Otherwise, plug the single USB/Keyboard adapter into your micro USB to USB adapter. This is just for setup—once your Raspbian installation has been configured to boot to the webpage, you won’t need a keyboard or mouse.
- Connect your Pi to the HDMI port of your HDMI display with an HDMI cable.
- Next, you will power on and boot the Raspberry Pi by plugging it in, so double check that everything else so far is plugged in and your display is set to the correct HDMI input.
- Plug in the micro USB AC power plug into the micro USB port on the Pi Zero labeled “PWR IN.”



- After a few seconds, you should see a series of Raspberry Pi load screens. It may take a few minutes for the Raspberry Pi to get set up this first time powering on, so don't panic if it takes longer than expected. When everything is finished, you should see a Windows/Mac-like desktop.

STEP 4: CONFIGURING RASPBIAN

- Open the Raspbian menu (click leftmost icon in the toolbar, which looks like a raspberry) and navigate to "Preferences," then "Raspberry Pi Configuration." A pop-up will open with multiple tabs.
- Raspbian's keyboard configuration defaults to UK settings. Navigate to the "Localisation" tab in the configuration window and select "Set Keyboard" to set your keyboard language.
- You may also wish to update your other localization settings at this point. Under "Localisation," you'll have option buttons to adjust your settings for locale (region language), time zone, keyboard, and WiFi country.
- In the configuration window, select the "System" tab and make sure the "Wait for network" option is checked. In the top right corner of your toolbar, click the network icon and select your wireless access point.

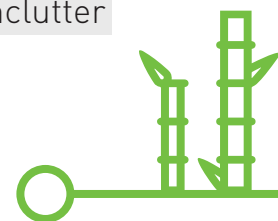
STEP 5: CONFIGURING AUTOSTART

- We're now going to use Raspbian's Terminal to add some settings to the autostart file. Start by clicking the terminal icon in the toolbar at the top of the screen.
- Type the following into the terminal: `sudo nano .config/lxsession/LXDE-pi/autostart`

PROTIP: You can use vi instead of nano, which is just another way to edit the file.

- Look for a line in the file that says `@xscreensaver -no-splash` and put a `#` in front of it, so the line reads `#@xscreensaver -no-splash`
- Next, you want to use the text editor in Terminal to add the following lines to the end of the file: `@/usr/bin/chromium-browser --kiosk --start-maximized --allow-running-insecure-content --remember-cert-error-decisions http://nationalzoo.si.edu/webcams/panda-cam @xset s off @xset s noblank @xset -dpms`
- Save your changes and close the file.

PROTIP: If you want to hide the mouse cursor, you can install unclutter with `sudo apt-get install -y unclutter` and add the following line to the autostart file: `@unclutter`



STEP 6: TESTING

- If you wish, you can now unplug the micro USB to USB adapter as well as the keyboard/mouse plugged into it.
- Your Raspberry Pi should now automatically boot to desktop, connect to your nearby WiFi, open the browser in fullscreen mode, and load the webcam page.

GENERAL TIPS

- Raspbian's keyboard configuration defaults to UK settings—depending on your keyboard, you may need to go to “Mouse and Keyboard Settings,” under “Preferences” in the Raspbian menu.
- Depending on your configuration, you may also need to change your WiFi settings for the U.S.—go to the Raspbian menu, then “Preferences,” then “Raspberry Pi Configuration.” A pop-up will open with multiple tabs. Under “Localisation,” you'll have option buttons to adjust your settings for locale (region language), Timezone, Keyboard, and WiFi Country.
- If you get really stuck, feel free to reflash Raspbian to reset everything.

SUPER SECRET TIP FOR EXPERTS

You can access your Raspbian installation over SSH from another computer on the same network. First, you need to turn on ssh access by adding a file named `ssh` to the root directory of your microSD card. Just `ssh pi@raspberrypi.local`. When you're prompted for a password, use `raspberry`, which is the default. (You should change this password if you don't want just anyone to be able to log in and mess with things.)

OTHER CAMS

All panda'd out? Here are some other Zipcar favorites:

[Dragon Pond Panda Cam \(Gengda, China\)](#)

[International Space Station Live Stream](#)

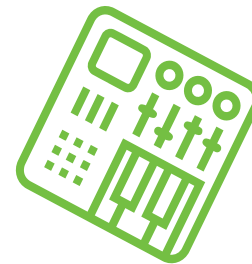
[Jackson Hole Town Square](#)



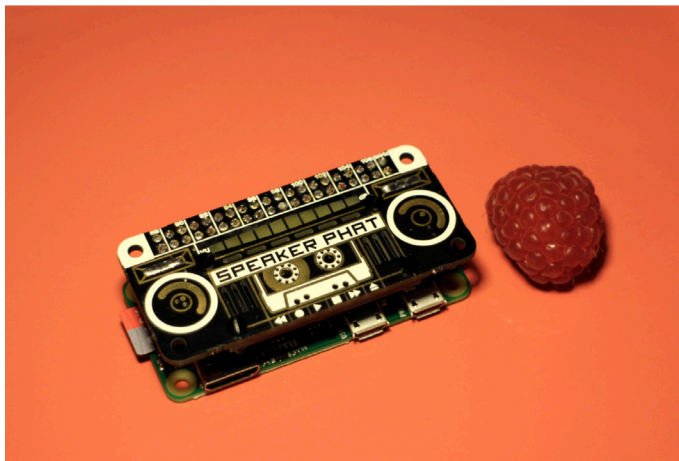
Lace up your boots,
print out a picture of Ada
Lovelace for inspiration,
and get to work!

MIDI SYNTH

by Alessandro Rossi



Have you ever heard of MIDI? The Musical Instrument Digital Interface is a well-established technology used to tell computers how to play sounds, and a staple in modern music production. MIDI controllers are simple pieces of equipment that can't produce sounds on their own, but are very expressive when coupled with music production software. In this guide, we are going to assemble a tiny software synthesizer using a Raspberry Pi Zero and few audio devices.



3-4 HOURS



ADVANCED

PHYSICAL INGREDIENTS:

- 1 Raspberry Pi Zero
- 1 HDMI cable
- 1 HDMI display (most modern televisions have an HDMI port)
- 1 mini HDMI to HDMI adapter
- 1 micro USB AC power plug
- 1 micro USB cable
- 1 micro USB to USB adapter
- A sound interface and speakers (or headphones). Here I'm going with the incredibly fun Pimoroni Speaker pHAT.
- A USB MIDI controller to play on. Any type you want, although keyboards, drum pads, and guitars work best for this guide.
- A USB to micro-USB adapter for your MIDI controller, unless it features a micro-USB plug already.
- If you can't connect your Pi Zero to the local network, you'll need a keyboard and monitor to work via command line. A modern TV makes a perfect monitor.

- A USB hub always comes in handy when you are dealing with multiple devices and you are constrained to a single port, but if the MIDI controller is your only USB device, then you are not going to need a hub.

OPTIONAL

- GPIO sound card (optional)
- Soldering iron and some solder
- A 2x20 0.1" male GPIO pins header, like [this one](#).

DIGITAL INGREDIENTS:

- [Raspbian](#)
- [Win32DiskImager](#) (for Windows)
- [ApplePi Baker](#) (for Mac)

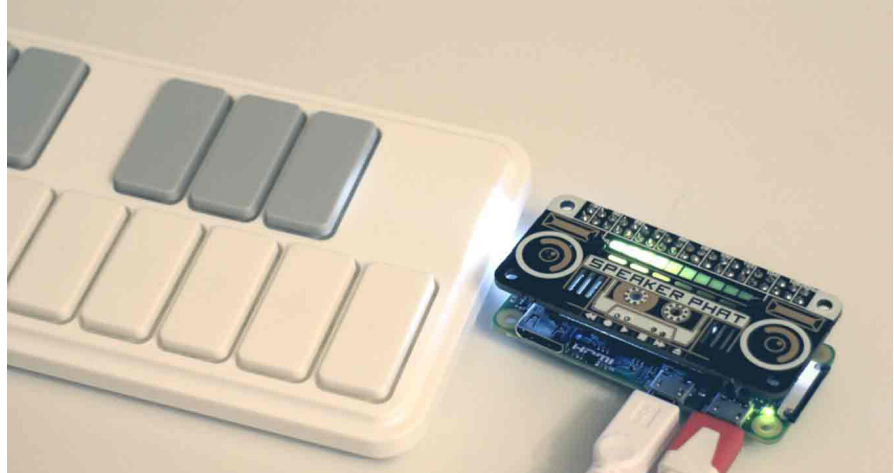
KNOWLEDGE NEEDED:

- How to download and manipulate files on your computer
- How to extract .zip files on your computer
- Configuration of your WiFi access point

THE SOUND INTERFACE

For the purpose of this guide, I'm going to be using the Speaker pHAT, which connects to the Raspberry Pi through the GPIO pins and communicates over the I2S bus. You don't actually need to know what all of that means, but if you are curious, there is tons of documentation on the [Raspberry Pi website](#). I chose the Speaker pHAT because it integrates the two main

components I'm going to need to produce sounds in a single board: a Digital-to-Analog audio Converter (DAC) and a speaker. Also, since it connects over GPIO, it leaves me free to use the one micro-USB port available on the Pi Zero, contributing to the overall lilliputian format.



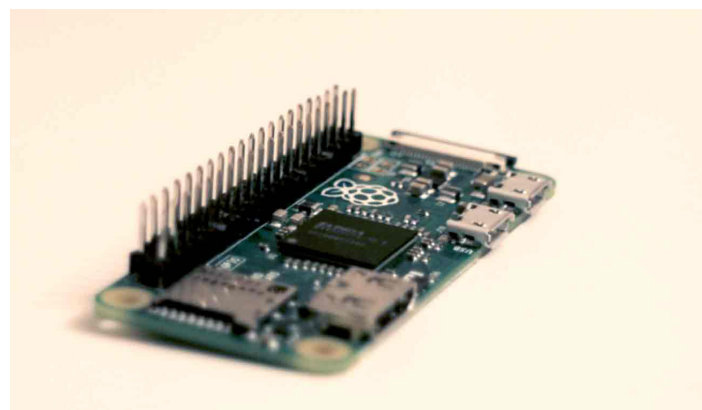
RECIPE



HARDWARE SETUP

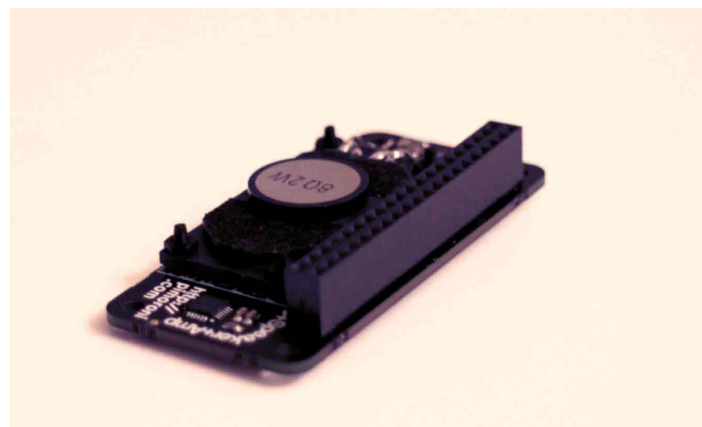
If you're going to go with the GPIO sound card approach, you're up for some soldering now. Those of you going for the USB audio solution instead may just plug in the sound card and skip this section.

Start with soldering the male GPIO pin header to the Raspberry Pi Zero, making sure that the pins point upward.



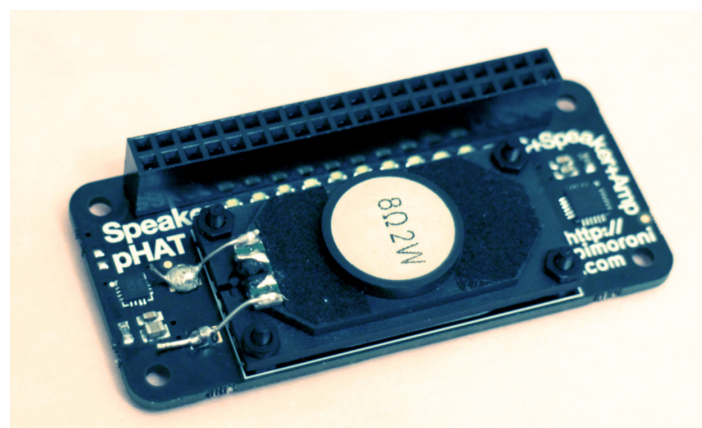
If you find it difficult to immobilize the pins while soldering, try wrapping some masking tape around the mounted header. Solder a few pins on both sides, then remove the tape and finish the work.

You'll then need to assemble your Speaker pHAT following the instructions provided by Pimoroni. Start by soldering the female GPIO header included in the kit in the same way you did for the male header, this time making sure that the socket is facing downwards.



Finish up by mounting the speaker and solder its contacts to the board.

Connect the pHAT to the Pi and pat yourself on the shoulder. You're done!



STEP 1: SYSTEM SETUP

Like any other Raspberry Pi projects, the first thing we need to do is writing the operating system image onto the SD card. So head over to raspberrypi.org and get the latest Raspbian OS. The lite version is all we need; a smaller system means more computational resources available to run the software synthesizer smoothly. Follow the latest installation guide for your OS on the Raspberry Pi website to learn how to prepare the micro SD card with Raspbian.

If you want to access your Zero over the network, once the card is ready find its boot partition, add an empty text file and name it `ssh`, with no extension. That will enable the SSH daemon and allow you to log into the Pi operating system using the `ssh` client or PuTTY. In order for the Zero to connect to the local WiFi, you'll need to provide a name and password for the network. In the same boot partition, create a `wpa_supplicant.conf` text file. Its content should look like this:

```
ctrl_interface=DIR=/var/run/wpa_supplicant GROUP=netdev
update_config=1

network={
  ssid="Your WiFi SSID"
  psk="your-password"
}
```

Set the SSID and password, watching for the correct letter case, and save the file.

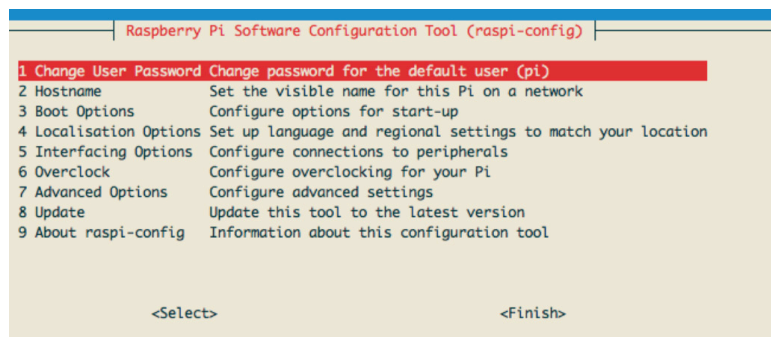
STEP TWO: FIRST ACCESS

Get that freshly baked SD card into the Pi, connect whatever is needed to get control over the running OS (keyboard and monitor or the network adapter), and finally plug in the power supply. The tiny green LED should light up, and if you've plugged into a monitor, you'll see the Linux system booting up. To access via SSH, wait a few minutes then try to ping `raspberrypi.local` from your command line. If the Pi connected to the network successfully, it'll reply to the ping and you'll get to know its IP address.

Once you see the login prompt, enter the default `user name - pi` - and `password - raspberry`.

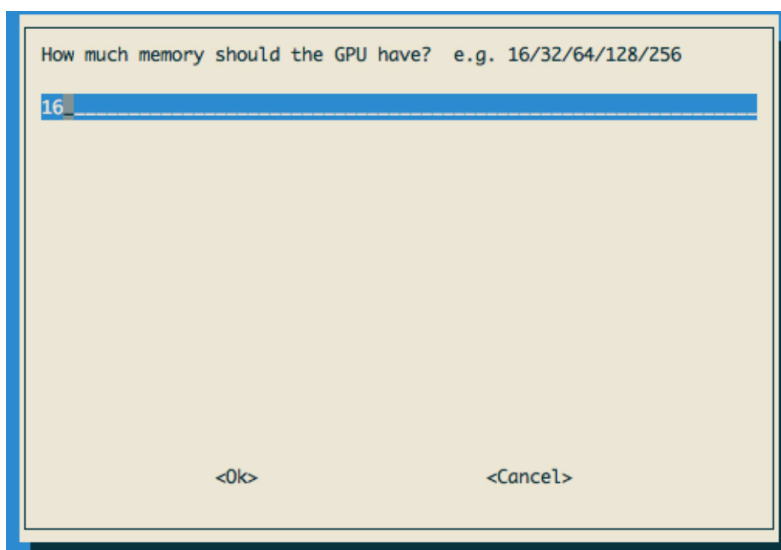
At the command prompt, run `sudo raspi-config`.

Raspi-config is a utility that ships with Raspbian OS which makes tweaking system settings very easy. Go ahead and change that password. You'll want to adjust a few more things before hitting the Finish button.



Localisation Options sets the default locale. It's not actually required to change anything here, but I noticed that installing new packages will show you a few warning messages if there isn't a default locale set. Take a minute to set it up.

Since we are going to run our Pi in headless mode, we can reallocate some of the GPU memory to be part of the system-wide pool. From the main raspi-config menu, go to Advanced Options, then Memory Split and dial the GPU memory down to 16mb.



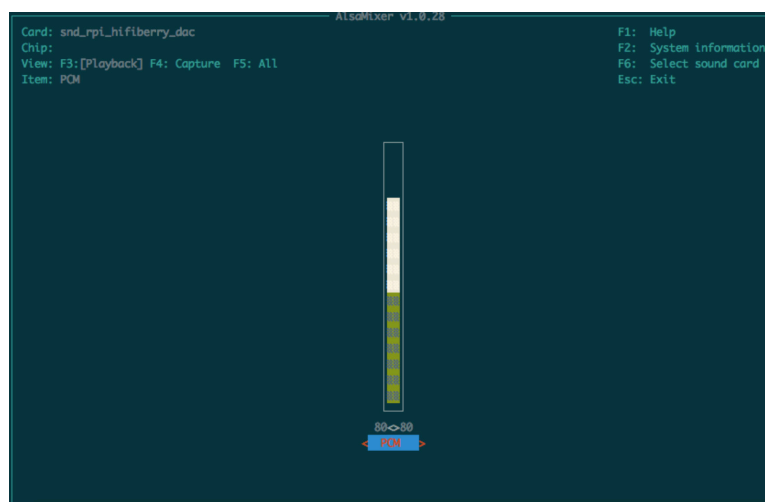
Hit Finish when done. You'll be asked to reboot.

Back at the login prompt, login again as the pi user, this time using the new password you just set.

STEP 3: INSTALLING THE SPEAKER PHAT DRIVERS

Pimoroni provides both drivers and installation instructions for the Speaker pHAT on their [github page](#). The setup script will take a few minutes to run and it will ask you to install a few additional packages along the way.

Wait for the setup process to complete and then check the settings of your sound card by running `alsamixer`. Turn up the volume a notch or two.



STEP 4: INSTALLING FLUIDSYNTH

Time to install the most important piece, the software synth. I chose fluidsynth for this project because it's open source, very simple to set up, and compatible with the SoundFont2 specifications.

This part is easy, good old apt-get: `sudo apt-get update && sudo apt-get install fluidsynth`

```
pi@raspberrypi:~$ sudo apt-get install fluidsynth
Reading package lists... Done
Building dependency tree
Reading state information... Done
The following extra packages will be installed:
  fluid-soundfont-gm fontconfig fontconfig-config fonts-dejavu-core libasound2 libaudio2 libavahi-client3 libcup
  libflac8 libfluidsynth1 libfontconfig1 libice6 libjack-jackd2-0 libjbig0 libjpeg8 liblcms2-2 libmmg1 libogg0
  libopus0 libpulse0 libqtcore4 libqtgui4 libsm6 libsndfile1 libtiff5 libvorbis0a libvorbisenc2 libx11-xcb1 libxi6
  libxrender1 libxt6 libxtst6 qsynth qtcore4-l10n x11-common
Suggested packages:
  fluid-soundfont-gs timidity nas cups-common jackd2 liblcms2-utils opus-tools pulseaudio libtapi0 qt4-qtdconfig jackd
The following NEW packages will be installed:
  fluid-soundfont-gm fluidsynth fontconfig fontconfig-config fonts-dejavu-core libasound2 libaudio2 libavahi-client3
  libcup
  libflac8 libfluidsynth1 libfontconfig1 libice6 libjack-jackd2-0 libjbig0 libjpeg8 liblcms2-2 libmmg1
  libogg0 libopus0 libpulse0 libqtcore4 libqtgui4 libsm6 libsndfile1 libtiff5 libvorbis0a libvorbisenc2 libx11-xcb1
  libxi6 libxrender1 libxt6 libxtst6 qsynth qtcore4-l10n x11-common
0 upgraded, 36 newly installed, 0 to remove and 13 not upgraded.
Need to get 129 MB of archives.
After this operation, 180 MB of additional disk space will be used.
Do you want to continue? [Y/n]
```

Fluidsynth ships with a default soundfont bank called `FluidR3_GM`, where GM stands for the [General MIDI](#) standard, and that gives you quite a few instruments to play with.

Launch fluidsynth with: `fluidsynth -a alsa -g 1 /usr/share/sounds/sf2/FluidR3_GM.sf2`
Bypassing those options, we're asking fluidsynth to use the ALSA audio driver, set the gain value to 1 (0 is silent, 10 is loud), and to load the GM sound bank. In addition to starting the synthesizer process, this command will also prompt you with the fluidsynth shell, which you can use to play the synthesizer programmatically. Ignore any warning messages you might get, for the moment. Feed the following line to the shell, then press enter: `> noteon 0 69 127`

Sweet! You hear that? We told fluidsynth to process the noteon signal, which is what MIDI controllers produce when you push the tone keys. The rest of the lines are its parameters.

0 is the MIDI channel. You have 15 of them, all allocated to the Grand Piano instrument by default. Type `inst 1` to see a list of all the instruments available in the sound bank, then `prog` to assign them to a channel. For instance, to get some synth drums on channel 1, type `prog 1 118`. 69 is the MIDI tone, a number used to encode specific sound frequencies (namely the musical notes) into a value that can be exchanged over the wire. 69 maps to the note A4 on your keyboard, 70 to Bb, 68 to G#, and so on, each number moving over the scale by a semitone. Finally 127 is the MIDI velocity, a parameter used to encode the energy applied on a key. Try the same command with a velocity of 50 for a softer sound. All those commands and parameters are documented on the fluidsynth [user manual](#).

Go on and play around with the shell. Once you're done, press `ctrl+c` or type `quit` to go back to the system prompt and move on to the next step.

STEP 5: HOOKING UP THE MIDI CONTROLLER

Go get your USB MIDI keyboard or drum pad and keep it handy. Type `aconnect -i` in the command prompt, and you'll get back the list of the input MIDI devices visible to the system. There is nothing interesting in that list just yet, but if you plug in your USB controller and re-type the same command, your controller should show up, looking somewhat like this:

```
client 20: 'nanoKEY2' [type=kernel] 0 'nanoKEY2 MIDI 1'
```

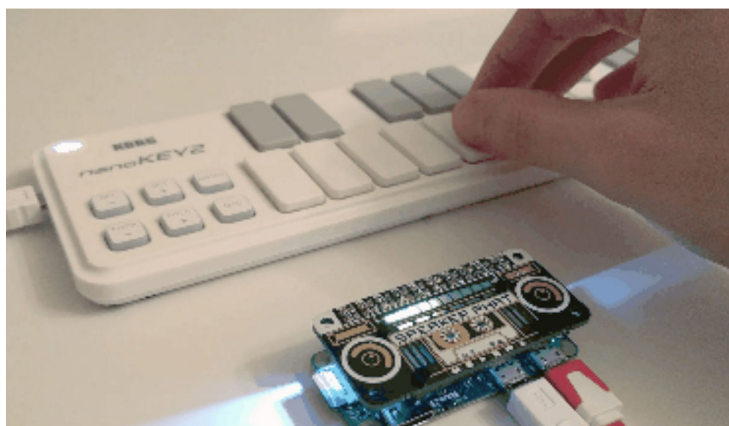
Now let's start `fluidsynth` again, this time passing a couple of new parameters: `fluidsynth -is -a alsa -g 1 /usr/share/sounds/sf2/FluidR3_GM.sf2 &` The `i` and `s` arguments tell `fluidsynth` to start in no shell and server mode, so we won't get a command prompt. In addition, we appended `&` at the end of the line to send the process to the background. Run `aconnect -o` to list the output devices. We expect `fluidsynth` to be one of them.

```
pi@raspberrypi:~ $ aconnect -i
client 0: 'System' [type=kernel]
  0 'Timer'
  1 'Announce'
client 14: 'Midi Through' [type=kernel]
  0 'Midi Through Port-0'
client 20: 'nanoKEY2' [type=kernel]
  0 'nanoKEY2 MIDI 1'
pi@raspberrypi:~ $ aconnect -o
client 14: 'Midi Through' [type=kernel]
  0 'Midi Through Port-0'
client 20: 'nanoKEY2' [type=kernel]
  0 'nanoKEY2 MIDI 1'
client 128: 'FLUID Synth (825)' [type=user]
  0 'Synth input port (825:0)'
pi@raspberrypi:~ $
```

Last thing left to do is telling ALSA to route any signals coming from the MIDI controller over to `fluidsynth`, basically the software equivalent of running a cable from your instrument to an amplifier. Take a look at the client names and numbers of both your MIDI controller and `fluidsynth`. In my case, my `nanoKEY2` controller is client number 20, port 0 and `fluidsynth` is client 128, port 0. To connect them together, I need to provide either their names or their numerical identifiers to the command `aconnect`, like so: `aconnect 20:0 128:0`

USB client identifiers may change at reboot or if you plug your device into different ports, but their name won't change. On the other hand, `fluidsynth` ID will most likely stay the same, but its name will change because it contains the current process ID number. Your best bet for an all-occasions command to connect your MIDI controller to `fluidsynth` looks like this: `aconnect 'midi name':0 ←fluidsynth-id→:0`

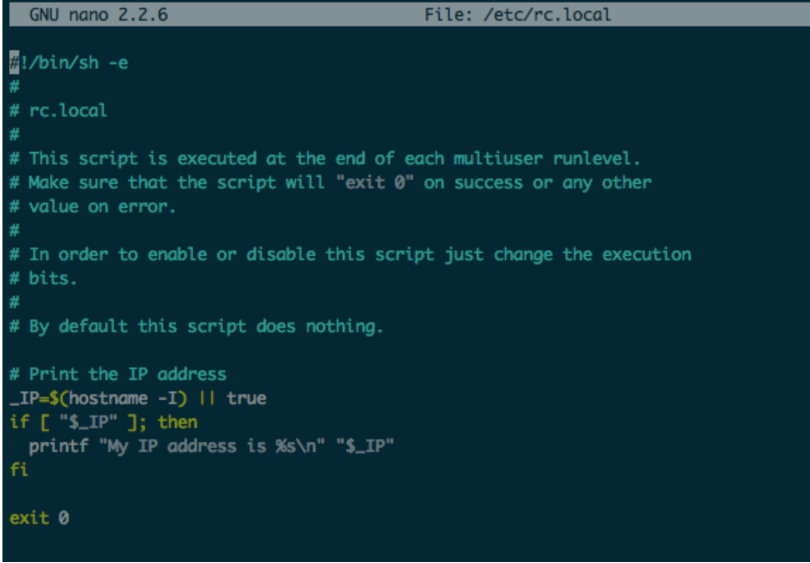
So in my case:
`aconnect 'nanoKEY2':0 128:0`



Alright! Time to play some tunes.

STEP 6: BOOTING UP AS A SYNTH

There's one last step left: getting rid of all these cables and adapters and configure the Pi to load fluidsynth every time we plug in the power. Linux systems have many different ways to achieve that, but there's an easy solution for Raspbian. `sudo nano /etc/rc.local`
`rc.local` is a script executed by the system at the end every boot cycle, perfect for those instructions we want to run only once at startup.



```
GNU nano 2.2.6 File: /etc/rc.local
#!/bin/sh -e
#
# rc.local
#
# This script is executed at the end of each multiuser runlevel.
# Make sure that the script will "exit 0" on success or any other
# value on error.
#
# In order to enable or disable this script just change the execution
# bits.
#
# By default this script does nothing.
#
# Print the IP address
_IP=$(hostname -I) || true
if [ "$_IP" ]; then
    printf "My IP address is %s\n" "$_IP"
fi
exit 0
```

Add these lines to the script, paying attention to leave `exit 0` as the last line of the file. It ensures that the system boots, despite any mistakes we might make with the rest of the instructions.

```
CONF_FILE=/home/pi/fluidsynth.conf
SOUNDBANK=/usr/share/sounds/sf2/FluidR3_GM.sf2
SYNTH_OPTIONS="-is -a alsa -z 1024 -c 8 -o audio.realtime-prio=95"
nice -19 fluidsynth $SYNTH_OPTIONS -f $CONF_FILE $SOUNDBANK &
sleep 10
aconect 'nanoKEY2':0 128:0
exit 0
```

Okay, let's try to break that down. Lines 1 through 3 are local variables. The first one refers a fluidsynth configuration file—we'll talk about it in a moment. Line 2 is the path to our GM sound bank. Line 3 is a set of fluidsynth options introducing a few new entries. `-z` and `-c` configure the audio buffer. You will need them only if you are using the Speaker pHAT and they should make any warning messages you received previously disappear. `-o` sets generic fluidsynth settings, in this case `audio.realtime-prio=95`, which grants higher execution priority to the audio thread, making the synth more responsive.

Line 4 in the script starts fluidsynth passing the content of the variables from line 1 through 3 as arguments. Also, fluidsynth is prepended with `nice -19`, another trick to improve the synth performances. `nice` is a unix command that alters the scheduled priority of a process. You can find more details on its main page, but fundamentally it will allow fluidsynth to use more CPU time than it would as a regular process.

Line 5 and 6 should be clear: Wait 10 seconds for the synth to start, then connect it to the MIDI controller. Remember to use your own MIDI client identifiers. Once you've finished editing your `c.local` file, press `ctrl+o` and `ctrl+x` to save and quit the nano editor.

Remember the configuration file we assigned to the `CONF_FILE` variable? That's fluidsynth's own startup script and we are going to use it to pre-load the synth channels with the configuration of instruments we like best.

At the terminal prompt, type `nano /home/pi/fluidsynth.conf` and write your own channel setup based on this example:

Any text after a `#` symbol is ignored, a convenient way to annotate the file. Here's a list of 'select' commands preloading my channels. The parameters are: channel, soundfont, bank, instrument

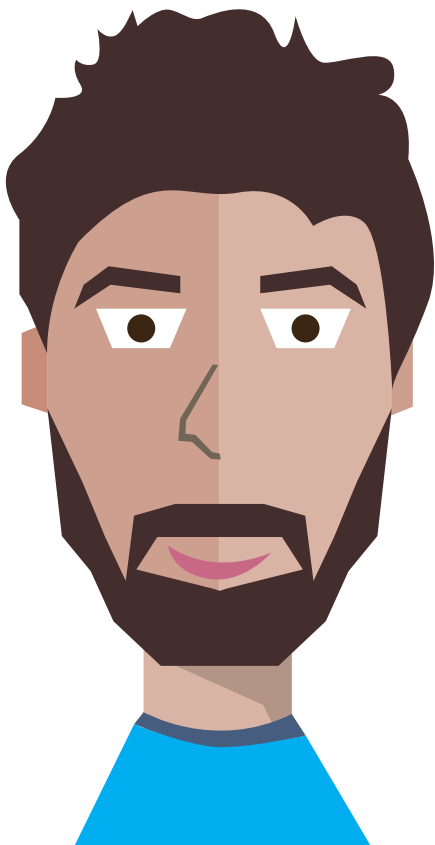
```
select 0 1 0 81 # sawtooth
synth select 1 1 8 24 # ukulele
select 2 1 0 2 # electric piano
```

STEP 7: SETTING THE GAIN

Set the gain as: `gain 1`



Save and exit nano, `sudo reboot` and enjoy!



Enjoy your
new super
compact MIDI
synthesizer!